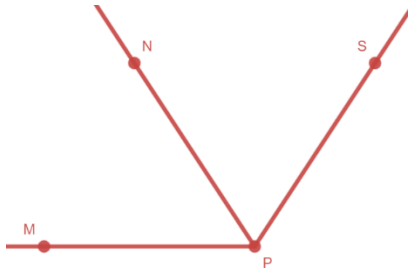


$\angle MPS$ is bisected by $\overrightarrow{PN}$. $m\angle MPN = (7x + 4)^\circ$ and $m\angle NPS = (16x - 68)^\circ$.

1. Draw a sketch that represents the description.



2. Find the value of x .

$$7x + 4 = 16x - 68$$

$$72 = 9x$$

$$x = 8$$

3. Determine $m\angle MPN$.

$$(7(8) + 4)^\circ = (56 + 4)^\circ = 60^\circ$$

4. Find $m\angle MPS$.

$$2(60^\circ) = 120^\circ$$

5. $\angle FBW$ is bisected by $\overrightarrow{BR}$ where the $m\angle FBW = (20x - 14)^\circ$ and $m\angle RBW = (\frac{5}{2}x + 8)^\circ$. Find $m\angle FBR$.

$$20x - 14 = 2\left(\frac{5}{2}x + 8\right)$$

$$20x - 14 = 5x + 16$$

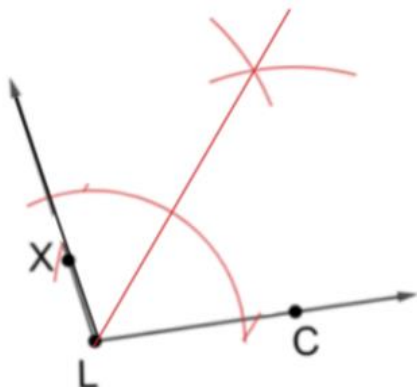
$$15x = 30$$

$$x = 2$$

$$\frac{5}{2}(2) + 8 = 5 + 8 = 13$$

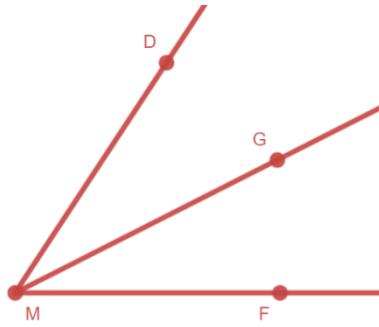
$$m\angle FBR = 13^\circ$$

6. Use a compass and a straightedge to bisect $\angle XLC$.



$\angle DMF$ is bisected by $\overrightarrow{MG}$ where $m\angle DMG = (4x)^\circ$ and $m\angle DMF = (2x + 24)^\circ$.

7. Sketch a diagram that could be used to represent this situation.



8. Find the value of x .

$$2x + 24 = 2(4x)$$

$$2x + 24 = 8x$$

$$24 = 6x$$

$$x = 4$$

9. Find $m\angle DMG$.

$$(4(4))^\circ = 16^\circ$$

10. Find $m\angle DMF$.

$$(2(4) + 24)^\circ = (8 + 24)^\circ = 32^\circ$$